
BIOLOGY

9700/43

Paper 4 A Level Structured Questions

October/November 2019

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **14** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
underline	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

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Question	Answer	Marks
1(a)(i)	Q – <u>receptor</u> ;	1
1(a)(ii)	R – proton / hydrogen ion / H^+ , pump / carrier ; S – calcium (ion) / Ca^{2+} , channel (protein) ;	2
1(a)(iii)	T – cell surface membrane / plasma membrane / plasmalemma ;	1
1(b)	<i>any five from:</i> 1 ABA / abscisic acid, binds / attaches, to receptor ; 2 calcium ions enter cell ; 3 (Ca^{2+} is) second messenger ; 4 proton / hydrogen ion / H^+, exit, stops / is inhibited ; 5 potassium ions / K^+, leave (cell) ; 6 water potential / ψ, increases (in, cell / cytoplasm) ; 7 water leaves (cell), by osmosis OR down, ψ / osmotic, gradient ; 8 cell, volume decreases / becomes flaccid / loses turgidity / loses turgor ;	5

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Question	Answer	Marks
2(a)	two homologous chromosomes circled ;	1
2(b)(i)	<i>any three from:</i> 1 same, size / length / number of (kilo)base pairs ; 2 same shape ; 3 same banding pattern ; 4 same (order of), genes / loci ; 5 they, pair up / form a bivalent, in, prophase 1 / meiosis 1 ;	3
2(b)(ii)	20 ;	1
2(c)(i)	<i>any two from:</i> 1 ref. to <u>reduction</u> division ; 2 halves the chromosome number ; 3 (so) fertilisation / fusion of gametes, gives, diploid / $2n$ / 40 ; 4 prevents chromosome <u>number</u> doubling (each generation) ;	2
2(c)(ii)	1 (by chance) both gametes (may) have recessive allele ; 2 zygote / offspring, has, pair / two / homozygous, <u>recessive</u> alleles ;	2
2(d)(i)	<i>any two from:</i> 1 (causes a) STOP codon ; 2 does not code for <u>amino acid</u> / no <u>amino acid</u> added to chain ; 3 stops <u>translation</u> OR rest of / later, codons / triplets / mRNA / sequence, not <u>translated</u> ;	2
2(d)(ii)	<i>parental genotypes</i> Aa X Aa and <i>gametes</i> A a A a ; <i>offspring genotypes</i> AA Aa (Aa) aa ; <i>offspring phenotypes</i> normal normal (normal) woolly ;	3

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Question	Answer	Marks
3(a)	$(10 - 5) / 5 \times 100$; 100 ;	2
3(b)	<u>restriction</u> , enzyme / endonuclease ; reverse transcriptase ;	2
3(c)	<i>any two from:</i> 1 small / low mass, so can enter, cells / <i>E. coli</i> (through membrane); 2 self-replicate in cell so multiply / make many copies of, gene ; 3 have restriction site(s) / can be cut by restriction enzymes, so new gene can, be added / join ; 4 have, marker / antibiotic resistance / fluorescence / reporter, <u>genes</u>, so, recombinants / transformed cells / cells that took up plasmid, can be recognised ; 5 have promoter so gene can be, expressed / transcribed ; 6 circular so, more stable / not damaged by host cell enzymes ;	2
3(d)	<i>any four from:</i> 1 prevents / less / lower, (named) disease / infection / pathogen transmission, risk ; 2 large / unlimited, supply OR mass production (from engineered bacteria) ; 3 cost of, purification / processing, lower ; 4 lower risk of, allergy / immune reaction / rejection / side effects ; 5 potential to, engineer / improve, recombinant proteins ;	4
3(e)(i)	fewer genes, expressed / have increased expression ;	1
3(e)(ii)	<i>any two from:</i> 1 transcription factors ; 2 (help / stop), binding / functioning, of RNA polymerase ; 3 ref. promoter ; 4 AVP ; light, is detected (by eyes) / causes changes melatonin enhancer / silencer, (DNA) sequences signalling, molecules / pathways	2

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Question	Answer	Marks
3(e)(iii)	<i>any two from:</i> 1 idea of compare with known, genes / sequences / genomes, in database ; 2 search / analysis, programme / software / algorithm ; 3 identify, role of protein / (named) health effects ;	2

Question	Answer	Marks
4(a)	<i>any two from:</i> 1 not, discrete / categorical / with only a few values ; 2 ref. to normal / bell(-shaped), distribution / curve ; 3 polygenic / many genes / multiple loci ; 4 environment has (large) effect ;	2
4(b)(i)	directional ;	1
4(b)(ii)	<i>any two from:</i> 1 (adults / penguins) <u>compete</u> for food ; 2 large bill size is a selective advantage OR those with large bills, get more food / get food more easily / survive ; 3 food availability is a <u>selection pressure</u> ;	2
4(c)	<i>any one from:</i> <i>males with bigger bills</i> 1 get more food for, offspring / chicks ; 2 can better defend, chicks / offspring ; 3 more chance of, getting / attracting, a, mate / female ;	1

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Question	Answer	Marks
5(a)	<i>any five from:</i> 1 cannot, exchange pollen / cross-pollinate / cross-breed / interbreed ; 2 <u>reproductively isolated</u> ; 3 because they live, too far apart / not near enough / at different altitudes ; 4 <u>geographical</u>, barrier / separation / isolation ; 5 because they have different pollinators ; 6 <i>ref. to</i> ecological isolation / different niches / adaptation / specialisation ; 7 colour significance explained ; 8 flower length / distance to nectar, significance explained ; 9 <i>ref. to</i> <u>pre-zygotic</u> ;	5
5(b)(i)	<i>any two from:</i> 1 same / equal, number of chromosomes ; 2 meiosis can occur in, hybrid / F1 / offspring ; 3 chromosomes (similar enough to) pair up ;	2
5(b)(ii)	<i>any two from:</i> 1 (F1 / hybrids have) fewer (F2) offspring / reduced (reproductive) fitness ; 2 outcompeted by, parent species / non-hybrids ; 3 parent, species / phenotypes, are, better adapted / more successful ; 4 <i>idea of</i> 'disruptive' selection ; 5 hybrid breakdown / hybrid line not sustained long-term ;	2

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Question	Answer	Marks
6(a)(i)	thylakoid (membranes) / lamella(e) / grana ;	1
6(a)(ii)	<i>any three from:</i> 1 ref. to LHC / light harvesting complex ; 2 (named) accessory pigments ; 3 pass, light / photons, to, reaction centre / chlorophyll <u>a</u> / primary pigment/chlorophyll P680 ; 4 <u>non-cyclic</u> photophosphorylation ; 5 more / different, wavelengths / energy frequencies, absorbed / used ;	3
6(b)	<i>any two from:</i> 1 as (concentration of) cadmium (ions) increases PSII <u>activity</u> decreases ; 2 supporting figures comparative quote with units ; 3 concentration rises by, order of magnitude / factor of 10, each time ;	2
6(c)(i)	<i>any four from:</i> - D1 in month A functional chlorophyll concentration is higher in unpolluted lake (than polluted lake) ; - D2 in month B functional chlorophyll concentration is same in both (lakes) ; - D3 in both (lakes) functional chlorophyll concentration is higher in month A than in month B ; - E4 in polluted lake copper ions, inhibit / damage / disrupt, chlorophyll ; - E5 in month B, protocists are, dormant / spores / resist entry of ions / AW ; - E6 month A =, summer / <u>hotter</u> / <u>higher</u> light intensity / <u>longer</u> days ; 7 AVP ; ref. flooding decreases Cu^{2+} concentration in month B evaporation increases Cu^{2+} concentration in month A	4
6(c)(ii)	substitute for, magnesium ion / Mg^{2+} ;	1
6(d)	<i>any four from:</i> 1 spot / extract, placed on, pencil line / base line / line of origin ; 2 repeat / concentrate, spot / extract ; 3 end / base, of chromatogram, suspended / placed, in solvent ; 4 R_f value = distance moved by, spot / pigment / solute ÷ distance moved by solvent (front) ; 5 compare with known R_f value to identify, chlorophyll a / pigment ; 6 detail of method ; e.g. cover to stop evaporation of solvent remove chromatogram before solvent front reaches top	4

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Question	Answer	Marks
7(a)	<i>any three from:</i> 1 <u>rate of amylase production</u> higher with gibberellin (compared to without) ; 2 with and without gibberellin figures ; <i>with gibberellin</i> 3 rate of amylase production increases, over time / for 13 hours OR amylase production gets faster, over time / till 13 hours OR rate increases more after 5 hours / rate is at maximum 13–15 hours / rate plateaus from 13–15 hours ; <i>without gibberellin</i> 4 rate of amylase production, is constant / does not change much / fluctuates in a narrow range / is low throughout ;	3
7(b)	<i>any three from:</i> 1 amylase enters endosperm ; 2 hydrolyses / breaks down / converts, starch / amylose / amylopectin ; 3 maltose / glucose, moves to / used by / needed by, embryo ; 4 (maltose / glucose) for respiration / to release energy / for ATP production ; 5 for growth of <u>embryo</u> ;	3
7(c)(i)	<i>Le</i> ;	1
7(c)(ii)	1 (dominant allele) codes for enzyme ; 2 converts inactive (gibberellin) to active gibberellin ;	2
7(c)(iii)	<i>any two from:</i> 1 cell division / mitosis ; 2 cell, elongation / enlargement ; 3 increase in internode length ;	2

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Question	Answer	Marks
8	1 glycolysis ; 2 cristae ; 3 ATP synth(et)ase ; 4 protons / hydrogen ions / H⁺ ; 5 matrix ; 6 Krebs / TCA / citric acid, cycle ;	6

Question	Answer	Marks
9(a)	<i>any eight from:</i> 1 not cellular ; 2 contain, nucleic acid / genetic material / DNA / RNA (core) ; 3 DNA may be single-stranded or double-stranded ; 4 (core surrounded by) protein coat / capsid / capsomeres ; 5 may have, external / lipoprotein, envelope / membrane ; 6 20–750 nm ; 7 obligate parasites ; 8 reproduced / replicated, in / by, host cells ; 9 disease-causing / pathogenic ; 10 no, metabolism / respiration / nutrition / excretion / growth ; 11 cannot move / immobile ; 12 have, proteins / enzymes, to help, infection / replication ; 13 (highly) specific to host (cells) ; 14 not (thought to be) living ; 15 AVP ; e.g. lytic / lysogenic, life cycles antigenic, variability / drift / shift	8

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Question	Answer	Marks
9(b)	<i>any seven from:</i> 1 name 1 ; 2 name 2 ; e.g. WWF / Greenpeace / Nature Conservancy / Wildlife Conservation Society / Oceana / Sea Shepherd / Conservation International / CITES / IUCN / IFAW / WAZA / World Seed Bank / IPBES 3 raise, funds / donations ; 4 influence, governments / businesses ; 5 ban / reduce, hunting / polluting / oil drilling / mining / deforestation ; 6 research / reports ; 7 conserve, species / populations / habitats / biodiversity OR prevent extinction ; 8 education / publicity campaigns / raise awareness ; 9 hold protests OR take direct action to prevent, development / exploitation ; 10 promote coexistence of wildlife and people ; 11 regulate / legislate for, trade in wild species ; 12 estimate / monitor / categorise, threatened / endangered, species ; 13 detail of CITES trade categories ;	7

Question	Answer	Marks
10(a)	<i>any seven from:</i> 1 action potential / depolarisation / impulse, at pre-synaptic <u>membrane</u> ; 2 (voltage-gated) calcium ion channels open / calcium ions enter (cell / cytoplasm / (motor) neurone / pre-synaptic knob) ; 3 vesicles fuse with pre-synaptic membrane ; 4 acetylcholine / ACh, released, by exocytosis / into synaptic cleft ; 5 (ACh) binds to receptors on, muscle cell membrane / sarcolemma / motor end plate ; 6 sodium ion channels open / sodium ions enter (muscle cell / sarcoplasm) ; 7 depolarisation of, (muscle) cell surface membrane / sarcolemma ; 8 (depolarisation) spreads / transmitted, to / down / via, T-tubules ; 9 depolarisation of (adjacent) sarcoplasmic reticulum (membrane) ; 10 (voltage-gated) calcium ion channels open ; 11 calcium ions, move / diffuse, out of SR / out of cisterna(e) ; 12 calcium ions, move / diffuse, into, sarcoplasm / cytoplasm ; 13 calcium ions, start contraction / bind to troponin ;	7

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Question	Answer	Marks
10(b)	<i>any eight from:</i> 1 mutation / allele / gene, on chromosome 4 / autosome ; 2 dominant ; 3 normal / recessive, allele has 10–35 repeats of CAG ; 4 HD / dominant / mutant, allele has, more / extra, repeats of CAG ; 5 larger number of repeats gives earlier onset ; 6 usual onset, after 28 / in middle age / before 65 ; 7 onset, in babies / from 1 year old, if very numerous repeats ; 8 ref. extra glutamine / polyglutamine ; 9 mis-folded, protein / huntingtin ; 10 neurological condition / brain problem OR (brain) neurones, die / destroyed ; 11 motor control uninhibited / involuntary movements / chorea ; 12 cognitive / mood, changes ; 13 AVP ; e.g. GABA producing neurones lost <i>ref. to basal ganglia / striatum, affected (first)</i>	8